

System Design & Requirements Specification (SRS)

Project Name: Entorno35

Version: 1.0

1. Introduction

1.1 Purpose

The purpose of this document is to define the technical and functional requirements for **Entorno35**, a cloud-native SaaS platform. The system facilitates compliance with **NOM-035-STPS-2018** for Mexican organizations by automating the distribution, scoring, and reporting of psychosocial risk assessments.

1.2 Scope

The system encompasses the complete compliance lifecycle:

1. **Multi-Tenant Administration:** Company registration, annual subscription management, and staff roster imports.
2. **Assessment Execution:** Secure, link-based delivery of **Guide I** (Trauma) and **Guide II/III** (Psychosocial Risk) questionnaires.
3. **Scoring Engine:** A bidirectional logic engine that processes weighted responses (0-4 vs. 4-0) according to question polarity.
4. **Intelligence Reporting:** Generation of individual legal PDFs and comprehensive "General Reports" aggregating workforce demographics and risk trends.

1.3 Definitions

- **ATS (Acontecimiento Traumático Severo):** Critical events (accidents, violence) requiring immediate medical referral.
- **Guide II:** Risk questionnaire for companies with 16-50 employees.
- **Guide III:** Risk questionnaire for companies with >50 employees (adds detailed domain evaluation).
- **Polarity:** The direction of scoring (Positive vs. Negative phrasing).

2. Technical Architecture & Design Patterns

2.1 High-Level Architecture

We will utilize a **Cloud-Native Microservices-Ready** approach deployed on Linux via Docker.

- **Frontend:** Next.js 14 (React) with Tailwind CSS/Ant Design.
- **Backend:** Go (Golang) using the Gin framework.
- **Database:** PostgreSQL (Relational integrity for strict hierarchies).
- **Infrastructure:** Docker Compose (Dev) / Docker Swarm or K8s (Prod).

2.2 Applied Design Patterns

To ensure code maintainability and scalability, the following patterns are mandated:

Pattern	Application Context
Strategy	Scoring Engine: Handles the two distinct evaluation methods. TraumaStrategy (Binary Yes/No) vs. RiskStrategy (Weighted Likert 0-4).
Composite	Result Aggregation: Treats individual Questions, Dimensions, Domains, and Categories as a unified tree structure to allow recursive score summation.
Observer	Async Stats: When a survey is submitted, an event triggers background workers to recalculate the company's "General Report" cache without blocking the user.
Factory	Report Generation: Dynamically assembles the correct PDF sections based on the active subscription year and company size (Guide II vs. III).

3. Functional Requirements

3.1 Module: User & Subscription Management

- **FR-SYS-01: Annual Cycle:** The system must enforce an annual testing window. New assessments cannot be started if the subscription for the current year is expired.
- **FR-SYS-02: Staff Rostering:** Tenant Admins must be able to bulk-upload staff via CSV including required metadata: Full Name, CURP, Role, Shift, and Department.

3.2 Module: The Assessment Engine

- **FR-ASM-01: Guide I (Trauma) Logic:**
 - Present binary "Yes/No" questions regarding severe events (accidents, assaults).
 - **Critical Trigger:** If specific "Yes" answers are detected (e.g., "Recuerdos persistentes"), the system must flag the profile as **"Requires Medical Attention"**.
- **FR-ASM-02: Guide II/III (Risk) Logic:**
 - Present Likert-scale questions (Siempre, Casi Siempre, Algunas Veces, Casi Nunca, Nunca).

- **Hierarchy:** Questions must be grouped visually by Category and Domain.

3.3 Module: Scoring & Logic (Core)

- **FR-CALC-01: Bidirectional Weighting:** The system must apply polarity rules before saving scores:
 - **Negative Questions:** (e.g., "Me preocupa sufrir un accidente") map **Siempre = 4** to **Nunca = 0**.
 - **Positive Questions:** (e.g., "Recibo capacitación útil") map **Siempre = 0** to **Nunca = 4**.
- **FR-CALC-02: Group Averaging:** For the General Report, the system must calculate the average score of all participants to determine the "Department Risk Level".

3.4 Module: Individual Reporting (PDF)

- **FR-RPT-01: Risk Visualization:** Render a linear meter chart showing the Total Score against the legal bands (Nulo <20, Bajo <45, Medio <70, Alto <90, Muy Alto >90).
- **FR-RPT-02: Action Plan Injection:** Automatically insert the legal "Necessity of Action" text corresponding to the final score.
 - *Example (Alto):* "Se requiere realizar un análisis de cada categoría... a través de un Programa de intervención".

3.5 Module: General Reporting (Company-Wide)

- **FR-GEN-01: Participation Stats:** Display a donut chart of (Completed / Total Staff) * 100.
- **FR-GEN-02: Demographic Profiles:** Aggregate and visualize staff distribution by:
 - Gender, Age Range, Marital Status, Education Level.
 - Time in Current Position, Shift Type, Shift Rotation, Total Work Experience.
- **FR-GEN-03: Global Risk Heatmap:** Display a matrix showing the **% of Staff** at each Risk Level (Nulo to Muy Alto) for every Domain (e.g., "Carga de trabajo").
- **FR-GEN-04: Item Analysis:** For each question, display a stacked bar chart showing the frequency of responses (e.g., "45% answered Siempre").

4. Database Schema Design (PostgreSQL)

Key Entities

1. **company:** id (UUID), rfc, subscription_end_date
2. **staff:** id, company_id, curp, demographics (JSONB)
3. **question_catalog:**
 - id, text
 - guide_type (I, II, III)
 - category_id, domain_id
 - polarity (POSITIVE / NEGATIVE)
4. **assessments:** id, staff_id, status, completed_at
5. **responses:**
 - assessment_id, question_id
 - selected_value (0-4)
 - score (Calculated based on polarity)

5. Non-Functional Requirements

- **NFR-SEC-01: Encryption:** PII (CURP, Name) must be encrypted at rest using AES-256.
 - **NFR-PERF-01: Scalability:** The system must support concurrent report generation for companies with up to 5,000 employees without timeout.
 - **NFR-LEG-01: Anonymity:** General Reports must automatically suppress demographic charts if the sample size is < 5 to prevent individual identification.
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6. Implementation Roadmap

1. **Phase 1 (Core):** Go API setup, PostgreSQL migration, and Seeder Script for the 72 NOM-035 questions.
2. **Phase 2 (Logic):** Implementation of RiskStrategy and the polarity inversion function.
3. **Phase 3 (Frontend):** Next.js dashboard for Assessment taking and Admin roster management.
4. **Phase 4 (Reporting):** Aggregation SQL queries for the General Report and PDF generation service.
5. **Phase 5 (Monetization):** Stripe integration for annual renewals.